Worksheet 37 · Bearings IGCSE

Edexcel IGCSE Mathematics · past paper questions

Name:	Subject: Math	Date:
Learning Objectives: Students will be able to: • Measure and write bearings as three-figure bearings • Find a back bearing from a given bearing • Use trigonometry to solve problems involving bearings		
Grade: / 34	**Teacher's Feedback:**	

Answer in the space given. Show your working. Marks for each part are shown in brackets.

Questions *Bearings*

1 The diagram shows the positions of two points, A and B.

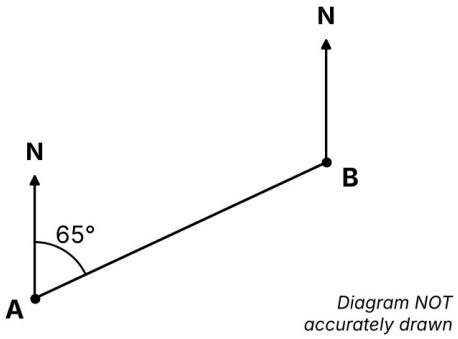


Diagram NOT accurately drawn

(a) Write down the bearing of B from A.

..... (1)

(b) Work out the bearing of A from B.

..... (2)

2 The bearing of Q from P is 130°.
Work out the bearing of P from Q.

..... (2)

3 The bearing of X from Y is 285°.
Work out the bearing of Y from X.

..... (2)

4 A ship sails 8 km due east from a port, P, and then 6 km due north to a point S.

(a) Work out the distance PS.

..... (2)

(b) Work out the bearing of S from P. Give your answer correct to the nearest degree.

..... (3)

5 A lighthouse, L, is 15 km from a port, P, on a bearing of 040°.
A buoy, B, is 9 km from P on a bearing of 130°.

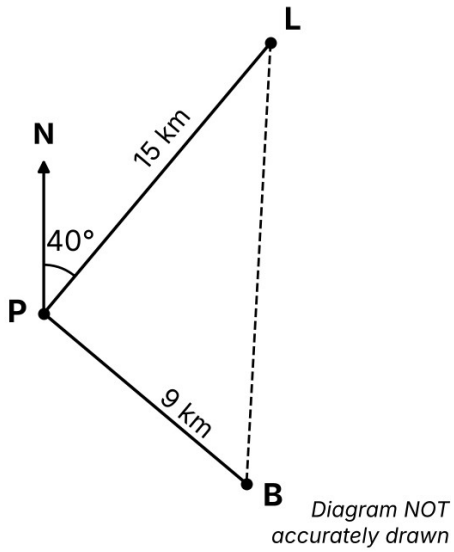


Diagram NOT accurately drawn

(a) Explain why angle LPB = 90°.

..... (1)

(b) Calculate the distance LB. Give your answer correct to 3 significant figures.

..... (2)

(c) Work out the bearing of L from B. Give your answer correct to the nearest degree.

..... (3)

6 Town C is due south of town A.
Town B is on a bearing of 120° from A and on a bearing of 060° from C.
Explain why triangle ABC is equilateral.

..... (3)

7 A plane flies 250 km from airport A to airport B on a bearing of 060°.
It then flies 400 km from B to airport C on a bearing of 150°.

(a) Show that angle ABC = 90°.

..... (2)

(b) Calculate the distance AC.

..... (2)

(c) Work out the bearing of C from A. Give your answer correct to the nearest degree.

..... (2)

8 A ship sails from a harbour, H, on a bearing of 035° for 18 km to a point A.
It then sails on a bearing of 110° for 25 km to a point B.

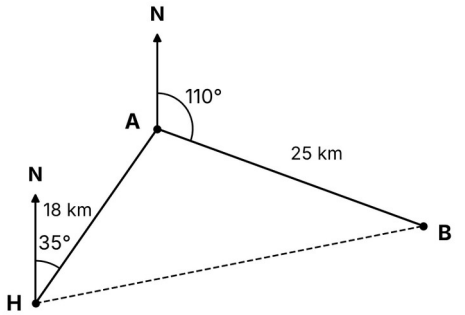


Diagram NOT accurately drawn

(a) Show that angle HAB = 105°.

..... (2)

(b) Calculate the distance HB. Give your answer correct to 3 significant figures.

..... (3)

(c) Work out the bearing of B from H. Give your answer correct to the nearest degree.

..... (2)